

Week 14

Noise Regulation and Planning

Spring 2021
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OCCUPATIONAL NOISE EXPOSURE

- **OSHA** (Occupational Safety & Health Administration) serves to ensure workplace health and safety
- It created the **OSH ACT** of 1970, whereby employers are responsible for providing a safe and healthful workplace
- Included in the act are *limits to noise exposure* to minimize or prevent hearing loss, and required actions when levels reach various criteria

OCCUPATIONAL NOISE EXPOSURE

- **Daily permissible dose** is a combination of level and time.
- As a starting point, for an **8-hour** workday, the maximum allowable continuous noise exposure is **90 dBA**
- For every increase in level of **5 dBA**, the time duration must be cut in half
- Exposure to impulsive or impact noise should not exceed **140 dB peak** sound pressure level (Linear level)
- Other countries can be more stringent, such as using 85 dBA as a starting point, and/or a 3 dB increase for cutting the duration in half.

Time duration per day (T) in hours	Maximum permissible exposure (dBA)
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
0.5	110
0.25 or less	115

OCCUPATIONAL NOISE EXPOSURE

- Dose can also be measured for workers to more accurately track all the changes in level throughout a workday
- Dosimeters are worn by the worker
 - Consist of a compact sound level meter that integrates the signal over time, and a readout device to give dose and peak levels
 - The microphone is typically located on the shoulder
 - Worn for the entire duration of the shift, which does not have to be exactly 8 hours (e.g., can include overtime, breaks, etc.)

DOSIMETER EXAMPLES:



Larson Davis Dosimeter



Bruel & Kjaer Dosimeter

OCCUPATIONAL NOISE EXPOSURE

CRITERIA:

- If the daily noise exposure $> 100\%$, feasible administrative or engineering controls shall be utilized
 - Administrative:
 - Reduce employee exposure by limiting time in noisy areas (e.g., rotate to different, quieter workstation)
 - Offer more breaks or reduce total work-shift time
 - Engineering Controls:
 - Reducing the noise at the source (e.g., quieter machines)
 - Reducing noise along the path (e.g., enclosures, partial barriers, room sound absorption, etc.)
- If such controls fail to reduce the dose $< 100\%$, personal protective equipment shall be provided and used to reduce sound levels within the levels of the table
- For noise exceeding 85 dBA over an 8 hour shift:
 - Hearing conservation program
 - Monitoring and sampling of employee exposures
 - Audiometric testing program – baseline and periodic testing

OCCUPATIONAL NOISE EXPOSURE

ENGINEERING CONTROLS:

- Reduce noise at the machine using isolation and damping for structureborne noise
- Enclose areas of the machine using effective barriers with acoustic lining
- Enclose entire machines, work stations, or section off building areas
 - Full enclosures with access panels
 - Partial and/or moveable barriers useful when machine operator is exposed to noise from other nearby loud machines

Example of barrier curtains:



Courtesy of Sound Seal

Barrier &
Quilted
Fiberglass
Absorber
Composites



NOISE ORDINANCES

- Local ordinances are generally of two types:
 1. **Nuisance ordinance** – often at the discretion of a police officer
 2. **Limitations** based on objectively measured noise
- Local ordinances and state/provincial laws can regulate any or all of the following
 - **Steady sound** (building equipment)
 - **Intermittent sound** (entertainment venues)
 - **Impulsive sound** (gun ranges)
 - **Tonal sound** (building equipment)
 - **Construction sound** (by time of day usually)
 - **Annoying sound** (dogs barking, broadcasts from vehicles, racetrack)
- Exclusions
 - **Trains and transportation vehicles** regulated by the state or federal government
 - **Emergency vehicles**

NOISE ORDINANCES

- Often written by legal departments and not noise experts
 - Limits, units, and measurement descriptions can be erroneous or unrealistic
- Intended to minimize complaints and litigation, but not a guarantee
 - Example:
 - In a rural area, the ambient levels can be very quiet. Noise from a source can be 10 dB above ambient and still not exceed ordinance noise limits.
 - The type of noise can vary greatly (e.g., hum, tonal, whine, whistle) and even though they may be below limits, they can be very annoying to residents trying to sleep.
 - In a area nearby a noise highway, the noise emitted from building equipment could exceed the ordinance limit, but still be below the average ambient noise levels.

Exterior Noise regulation

- The FAA (Federal Aviation Administration), HUD (Department of Housing & Urban Development), the FTA (Federal Transit Administration), the FHWA (Federal Highway Administration) and some localities have exterior noise guidelines and regulations.

HUD Standard for New Housing

Site Classification	L_{dn} (dBA)
Acceptable	≤ 65
Normally acceptable	65 - 75
Unacceptable	> 75

Noise ordinances

- Example of local noise ordinance:
 - Excerpt from Livonia, MI, noise ordinance:
 - Note maximum noise limit varies by receiving land use

8.32.080 Maximum permissible sound levels by receiving land use.

- A. Except as otherwise provided by this chapter, on or after June 1, 1988, no person shall operate or cause to be operated on private property any source of sound in such a manner as to create a sound level which exceeds the limits set forth for the receiving land use category in Table I when measured at or within the property boundary of the receiving land use.

Table 1. Sound Levels by Receiving Land Use

Receiving Land Use Category	Time	Sound Level Limit dB(A)
Residential area	7:00 a.m.—11:00 p.m.	55
	11:00 p.m.—7:00 a.m.	48
Commercial area	At all times	65
Industrial area	At all times	75

Noise ordinances

- Example of local noise ordinance:
 - Excerpt from Southfield, MI, noise ordinance:
 - Note units are (erroneously?) listed as dB, not dBA

- (3) Noise: The intensity level of the sounds being emitted from any operation within the TR District shall not exceed the following decibel levels when adjacent to the following types of use districts:

<u>Sound Levels in Decibels</u>	<u>Adjacent Use</u>	<u>Where Measured</u>
50	All Residential Districts	Common Property Line
55	Neighborhood Business District	Common Property Line
60	General Business District	Common Property Line
65	Light Industrial District	Common Property Line
75	Heavy Industrial District	Common Property Line
75	Major Street	At Street Line
60	Secondary Residential Street	At Street Line (Residential Opposite)

Objectionable noises, due to intermittence, heat frequency or shrillness shall be muffled so as not to become a nuisance to adjacent uses.

Method of Measurement: The sound level, in decibels, measured at the points adjacent to the various use districts shall meet the established levels ninety (90%) percent of the time, and shall at no time exceed the established sound levels by more than ten (10%) percent; the sound levels shall be measured with an approved standard sound-level meter incorporating a 40 db. frequency-weighting network, or by any method subsequently approved by the U. S. Bureau of Standards.

Interior Noise Legislation: HUD

- US Dept. of Housing and Urban Development (HUD) requirements haven't changed in many years
- Building code is regularly updated

Barrier	HUD Recommendations [8.5]			Building code requirements
	Type of dwelling units			
	Grade I	Grade II	Grade III	
Partition wall	STC $\geq$ 55	STC $\geq$ 52	STC $\geq$ 48	STC $\geq$ 50
Floor ceiling assembly	STC $\geq$ 55 IIC $\geq$ 55	STC $\geq$ 52 IIC $\geq$ 52	STC $\geq$ 48 IIC $\geq$ 48	STC $\geq$ 50 IIC $\geq$ 50

Week 14

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Control Mechanisms for Noise



- It is most effective to control noise at its *source*.
 - Buy/make/design quiet sources
- Control the transmission *path* of airborne sound
 - Separate noisy spaces from noise sensitive areas
 - Sound blocking constructions
 - Add sound absorptive materials in source and/or receiving rooms
- Control at the *receiver*
 - Hearing protection

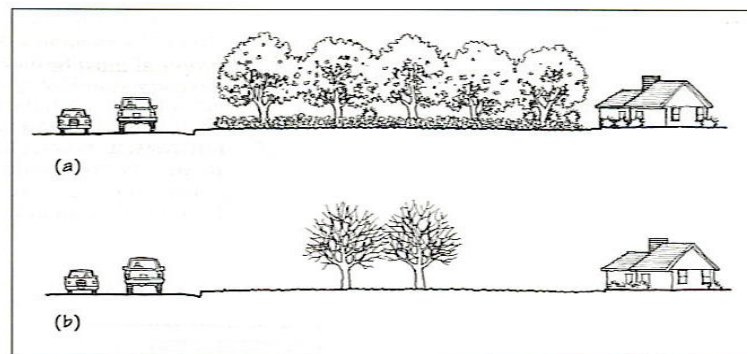
Site selection

LOCATION

- Selecting an appropriate site is key to minimize expenses paid for noise and vibration control
 1. Identify and measure noise sources that could propagate and intrude into your site/building
 2. Determine if noise and vibration sources from your site could create a disturbance in the community
 - Cost of noise controls
 - Nearby types of land use (residential, commercial, industrial, etc.)
 - Noise ordinances / potential litigation
 3. For vibration sensitive applications, identify potential sources (e.g., railroad, roadways, nearby industrial buildings, etc.) and measure ground vibration

Site Planning

- The larger the distance between the source of the noise and the receiver, the lower the noise level.
- Wide bands of foliage and trees absorb some sound.
- Natural shielding (e.g., berms) can reduce the need for noise control
 - 10 dBA of attenuation for a band of dense plantings of trees and shrubs 100 ft or more wide
 - Mature evergreen vegetation (>20 ft wide) may provide 2 to 4 dBA if the belt is sufficiently high and long with dense foliage extending to the ground

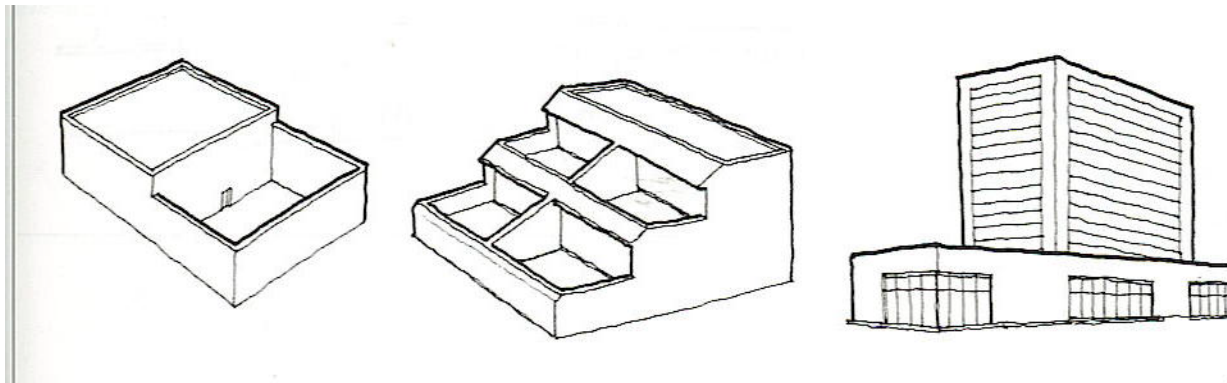
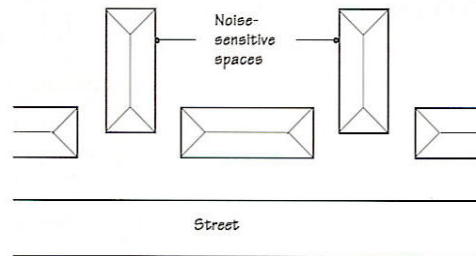


8.12 (a) Several rows of trees with heavy foliage provide attenuation of sound in addition to that obtained due to the distance effect.

(b) One or two rows of trees with little or no ground foliage give no additional sound attenuation except that due to the distance.

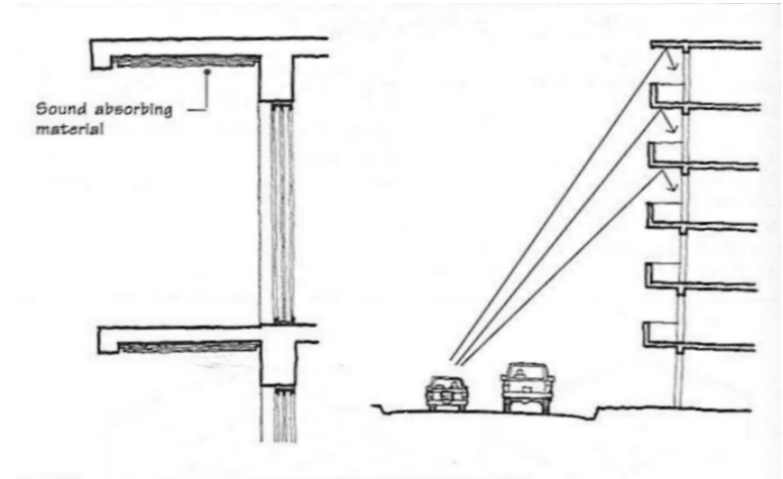
Site Planning

- Orienting building perpendicular to the street helps to minimize the effects of traffic noise for sensitive areas.
- Buildings (or building sections) can be used to block noise from sensitive areas or other buildings.



Site Planning

- The addition of **sound absorbing material on the underside of an overhanging balcony** helps to keep street noise from being reflected toward an occupied space.
- **Segregate areas need quiet** (e.g., residential areas, hospital zones, etc.) from noisy ones (e.g., industrial areas, ballparks, etc.).
- **Traffic should be concentrated on a few major streets** rather than spread out on many streets.



Interior Noise Control:

Space Planning

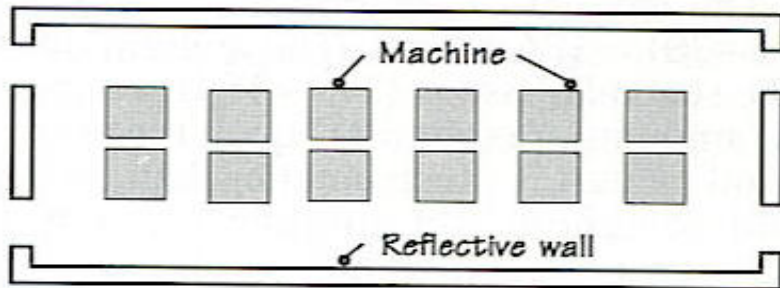
Architecture design can have a great impact on sound control.

- Separate noisy (or high vibration) rooms in plan *and* elevation from those requiring quiet through distance and buffer zones.
 - This method is *almost always the most inexpensive noise control technique*.
- Group noisy spaces, such as stacked mechanical or toilet rooms, unless the noisy spaces themselves need to be separated from one another, such as in music spaces.

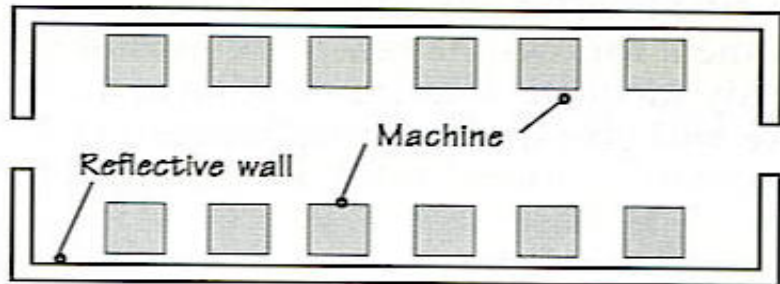
Interior Noise Control:

Space Planning

- Locate noise sources away from walls to reduce sound reflectance (reverberation), as shown in the manufacturing room (left below).



Preferred



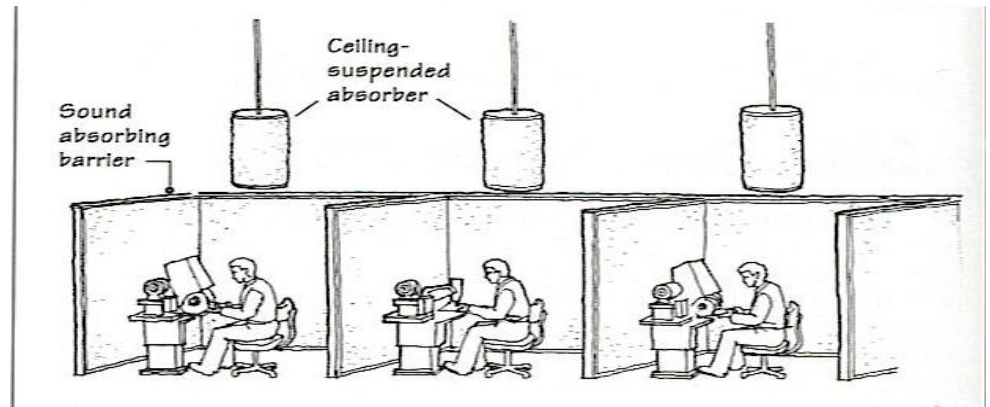
Not as Effective

Interior Noise Control:

Space Planning

Use multiple strategies

- BARRIERS
 - ABSORBERS
 - ISOLATORS
 - DAMPERS
- Example: Both ceiling hanging absorbers plus sound absorbing and blocking partial height barriers are used to reduce sound by placing them **close to the source** of the noise.



ARC4443 Acoustical, Electrical, and Illumination systems

Week 16

Acoustic Application

ACOUSTICS APPLICATIONS

- Speech/presentation auditoria, lecture hall, courtroom
- Drama Theater
- Concert Hall
- Multi-purpose Theater
- Support spaces
 - Music practice
- Cinema
- Other spaces: Offices, gathering spaces, health care, libraries

SPEECH PRESENTATION

- Speech/presentation auditoria, lecture hall, courtroom
- Speech intelligibility is maximized by increasing the signal-to-noise ratio (maximizing loudness of speech and minimizing ambient noise level)
- Five design considerations to achieve this
 - Optimize reverberation time
 - Eliminate acoustical defects such as echoes and flutter echoes
 - Maximize loudness in audience
 - Minimizing the ambient noise level
 - Provide speech reinforcement system where needed

SPEAKER-LISTENER DISTANCE

Design goal: minimize speaker to listener distance

- Speech intelligibility decreases rapidly with distance
- Seeing the speaker enhances intelligibility
 - Listener is deprived of direct sound if speaker is not seen
 - Direct sound helps localize speaker
 - Seeing facial expressions, gestures, body movements and lip movements enhances intelligibility

SPEAKER-LISTENER DISTANCE

- Maximum suggested design distances between stage and furthest seat
 - Lecture hall: 80 ft
 - Auditorium: 80 ft
 - Drama theater: 65 ft

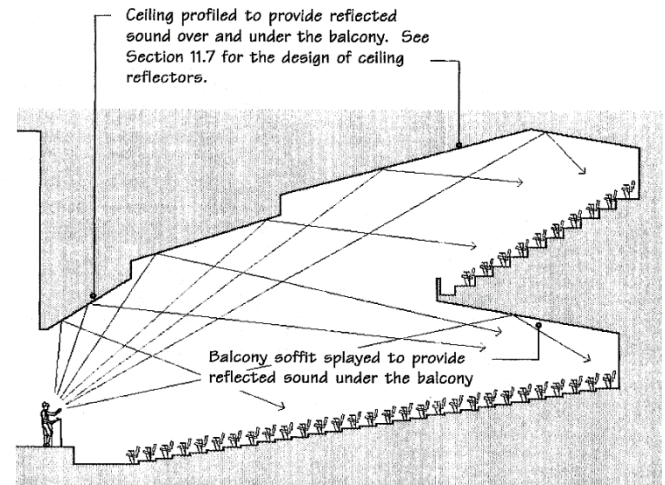
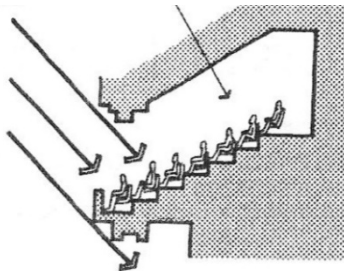
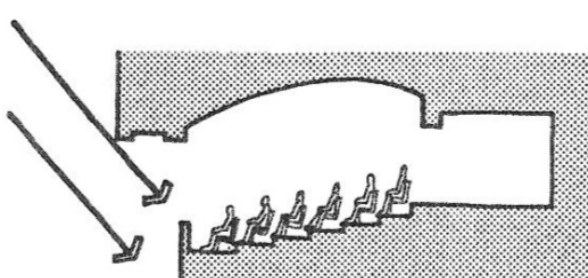
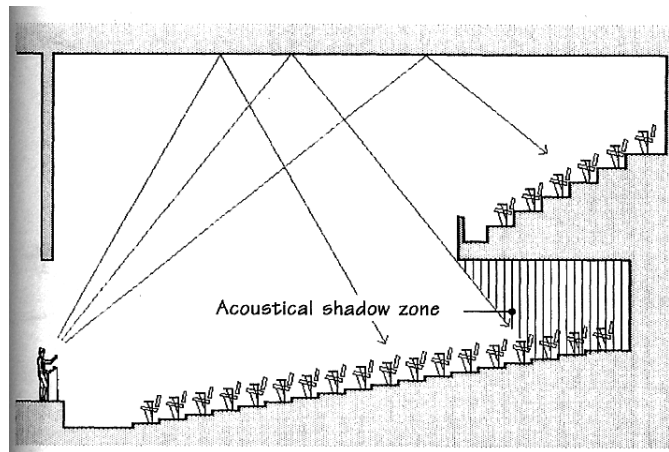
Table 11.1 Speech-supporting Elements and the Maximum Distance of Their Visibility

Visibility of speech-supporting element	Maximum distance	
	(m)	(ft)
Facial expressions	12	40
Gestures	20	65
Large body movements	30	100

Adapted from Reference 11.4.

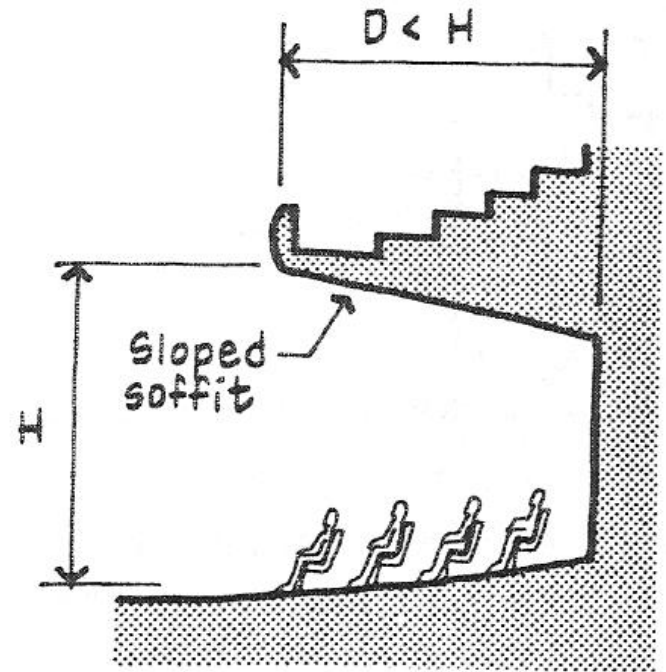
BALCONY AND HALL DEPTH

- Balconies are typically used when **audience exceeds 800** to control speaker-listener distance
- **Avoid acoustical shadows** due to deep and improperly shaped balconies and ceiling reflectors – *use ray analysis*



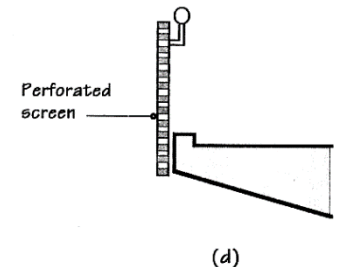
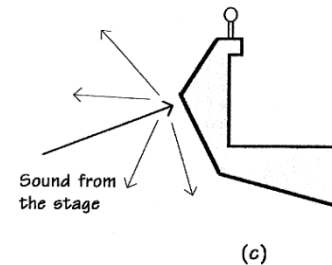
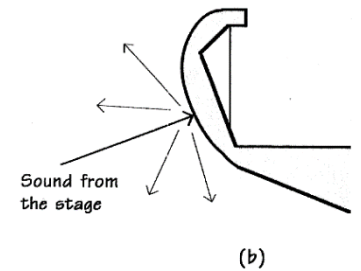
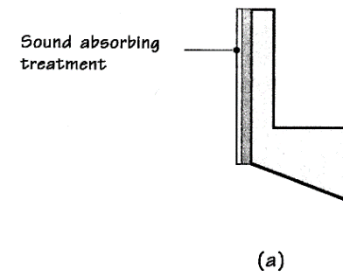
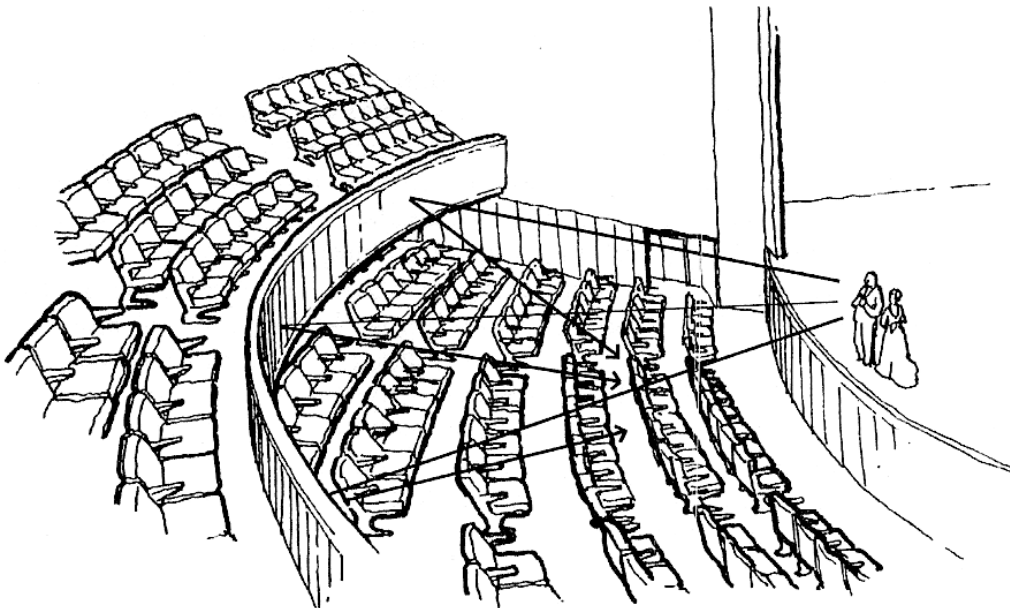
BALCONY AND HALL DEPTH

- Rough guide: Balcony overhang (D) should be no more than 2 times the height of balcony opening (H) in speech halls
- Cinema – $D \leq 3H$
(2H is preferred)
- Opera halls – $D \leq 1.5H$
- For concert halls – $D \leq H$
- Make sure last seat has a line of sight to the loudspeaker cluster or front third of ceiling



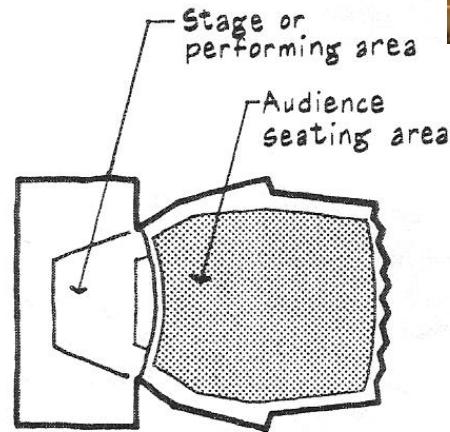
BALCONY AND HALL DEPTH

- Balcony parapet can produce unwanted reflections and echoes, so **acoustically treat vertical surface of parapet** that faces the stage.
- *Options: absorptive, diffusive or transparent*

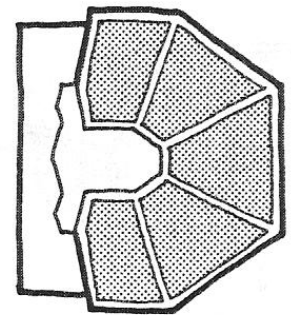


ROOM SHAPE

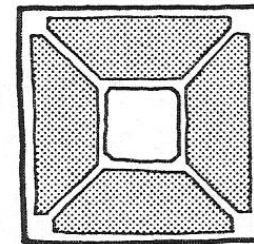
- **Proscenium stage** – stage house viewed through arch or rectangular opening
- **Open or thrust stage** – performing area extends into audience
- **Arena stage/stage-in-the-round** – audience surrounds the stage
- Extra special attention to reflecting surfaces needed for open and arena stages



Proscenium



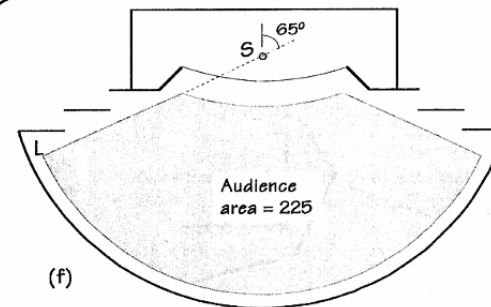
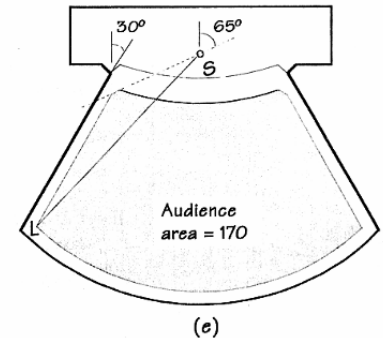
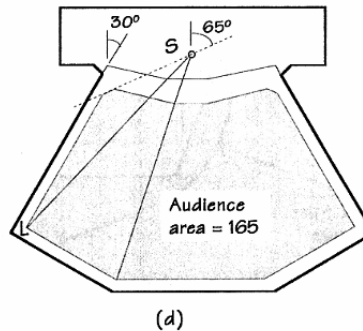
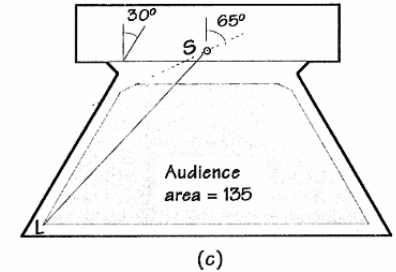
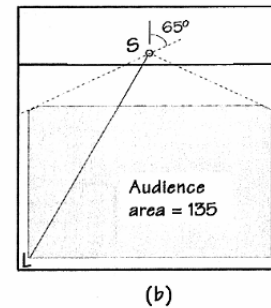
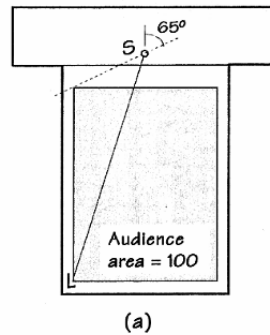
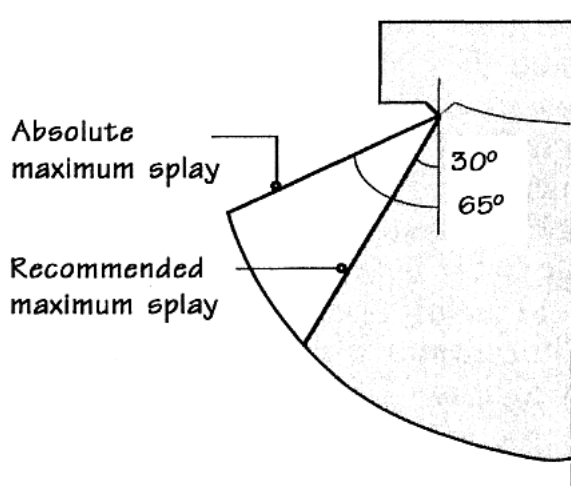
Open or Thrust



Arena

ROOM SHAPE

- Fan shape room minimizes speaker-listener distance
- Considering directionality of speech, a max. side-wall splay of 30° is suggested & not to exceed 65° splay



ROOM VOLUME

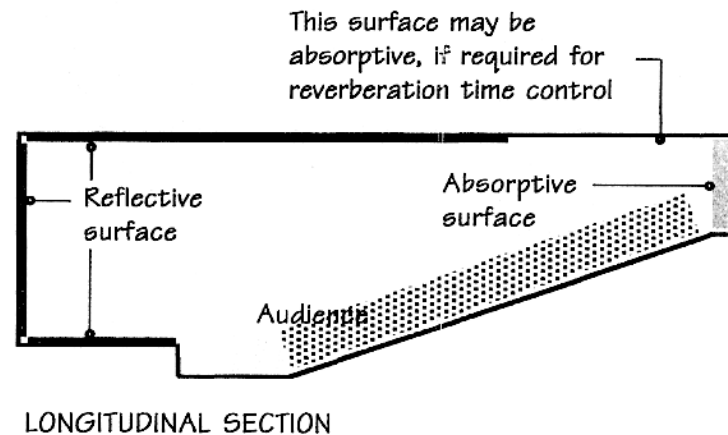
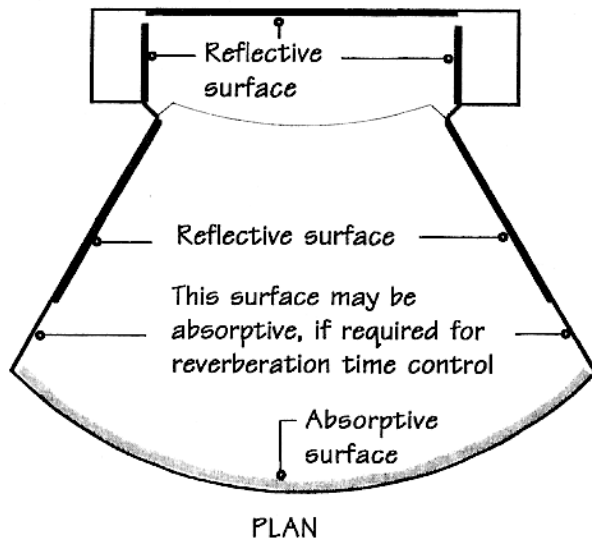
- For speech, minimizing room volume is desirable
 - More sound energy per seat
 - Less absorption is necessary to achieve recommended reverberation time with a smaller volume. This results in a greater proportion of reflective surfaces which in turn boosts loudness.
- * *Larger halls have greater ceiling height, which results in a greater volume per seat.*

Table 11.2 Recommended Floor Area and Room Volume per Seat

Area/seat	0.55 - 0.7 m ² (6.0 - 7.5 ft ²)
Volume/seat (classroom, lecture hall and drama theater)	2.0 - 5.0 m ³ (70 - 175 ft ³)

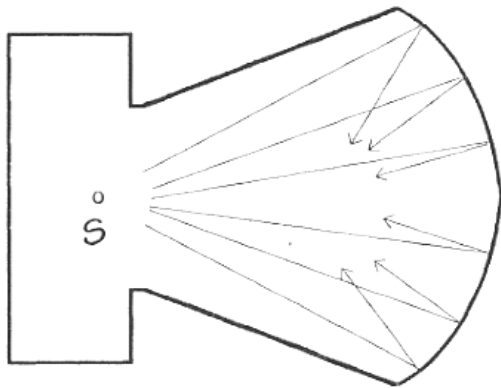
REFLECTING AND ABSORBING PARTS OF A ROOM

- Typically, **most absorption in a hall is provided by the audience**
- Generally, **surfaces near the speaker should be reflective** to support projection of sound.
 - Reflective surfaces produce early reflections which are integrated into direct sound if delay is 50-60 milliseconds or less
 - Reflective surfaces on stage provide feedback so speakers and actors can hear themselves

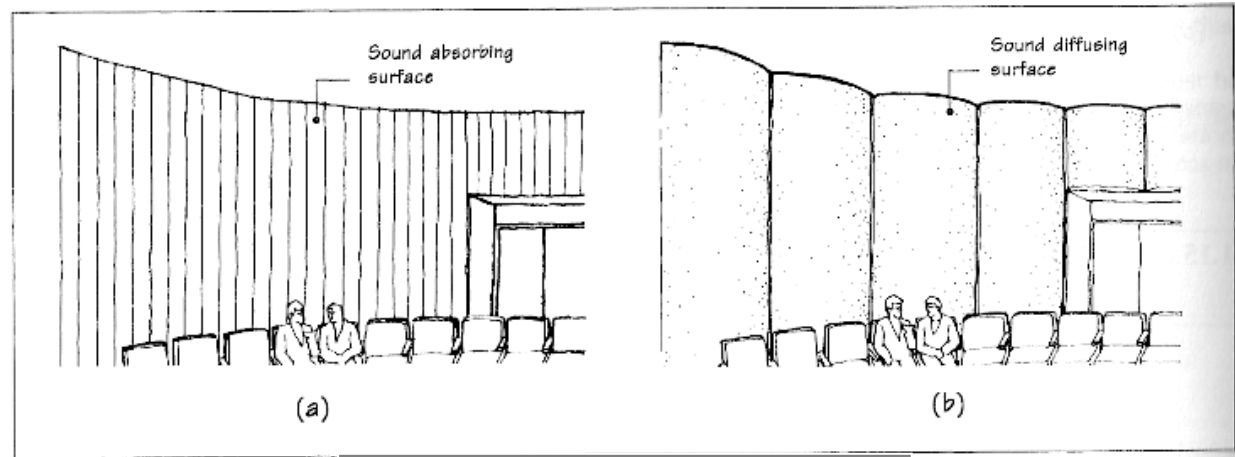


REFLECTING AND ABSORBING PARTS OF A ROOM

- Sound absorption or diffusion should be placed in the rear of the hall *to reduce echoes and control reverberation*
 - Large and medium sized halls have the potential for echoes from rear wall (if the difference between path length of direct sound minus the reflected sound is greater than 65 feet)
 - Generally, treat rear wall with sound absorbing or diffusing elements except for small rooms (100-150 people)



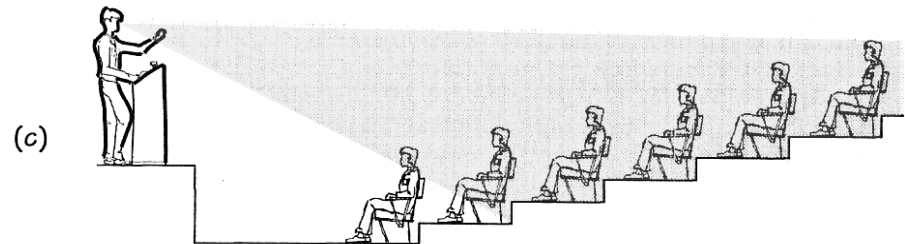
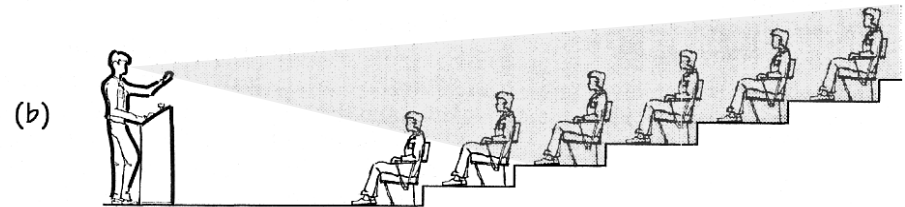
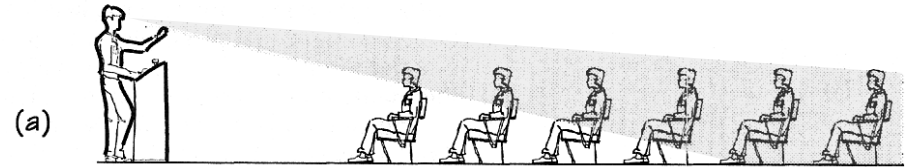
Avoid!



Mehta Fig. 11.17 p

FLOOR RAKE

- Floor rake is acoustically desirable
 - Improves amount of energy reaching audience
 - Reduces audience absorption as sound travels over audience at grazing incidence. Audience (a) is more absorptive than (c)

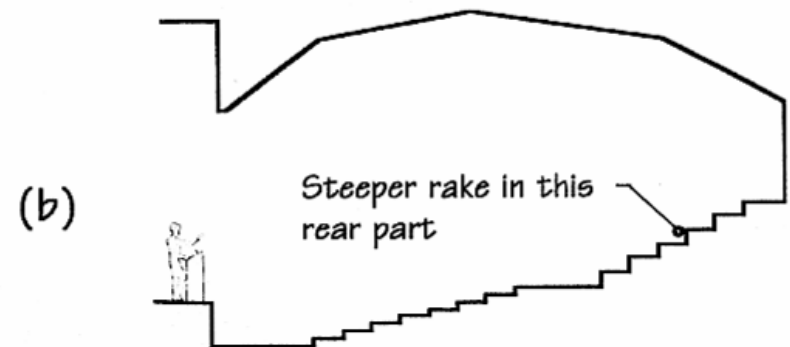
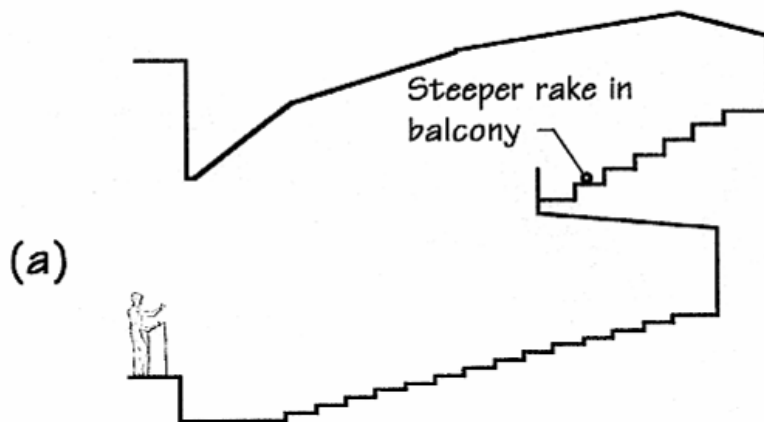


FLOOR RAKE

- Flat floors are possible for small audiences that have a *clear sight line* to speaker
 - *Sight line* – a line that joins the eye of the seated person with the focal point
 - *Focal point* – lowest point in the stage area which must be visible to every person
 - *Clear sight line* – no visual obstruction in sight line when the sight line clears over the head of the person in front
- Flat floors can only be used if the focal point is higher than 1.1 m (3.6 ft)

FLOOR RAKE

- Sloped floor usually required except in small halls
- Balcony floor slope must be greater than the main floor to achieve clear sight lines
- Seating behind a mid floor landing must have a greater floor slope than the seating in front

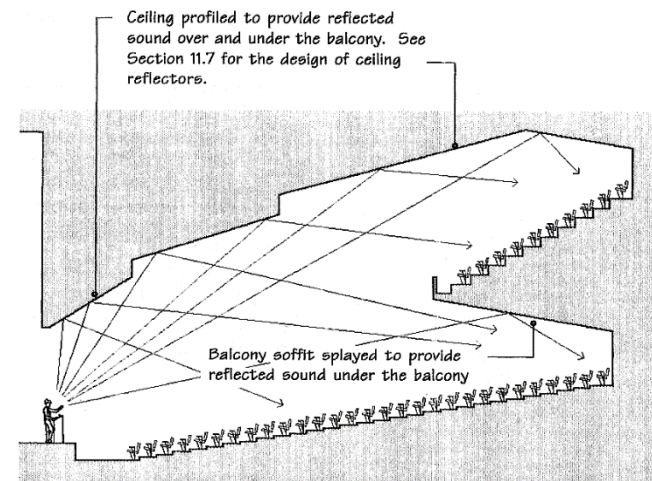


STAGE FLOOR

- Stage height should not be raised so much that front row cannot see stage floor (approx. 3' -6" max)
- Lecture halls: stage surface can be any reflective material
- Theatrical and musical use: construct stage floor of wood planks over an air space at least 8" deep
 - Air space provides some amplification of low frequencies
 - Comfort and safety of dancers (resiliently mounted wooden floor even better – true for dance studios also)

CEILING REFLECTIONS

- **Overhead reflections** from the ceiling seem to come from speaker on stage since our ears cannot localize sounds in the vertical direction
- **Large spaces need the ceiling profiled** to direct as much sound as possible **to the rear portion of hall**
- Ceiling reflectors may be a continuous or segmented ceiling
 - *Suspended panels allow easier access to building services and provide gaps for lighting and HVAC*



CEILING REFLECTIONS

- Reflector profile is determined by geometrical acoustics
- Large halls use multiple reflectors. Small halls may only use one or two reflectors at the front.
- Convex reflectors and surface irregularities diffuse sound and reduce acoustical glare
 - *Acoustical glare - excessive high frequency spectral reflections*
- For speech, the reflector size must be at least 8 ft in any one direction
- Must be heavy to efficiently reflect sound and to lower the reflector's resonant frequency

AMBIENT NOISE

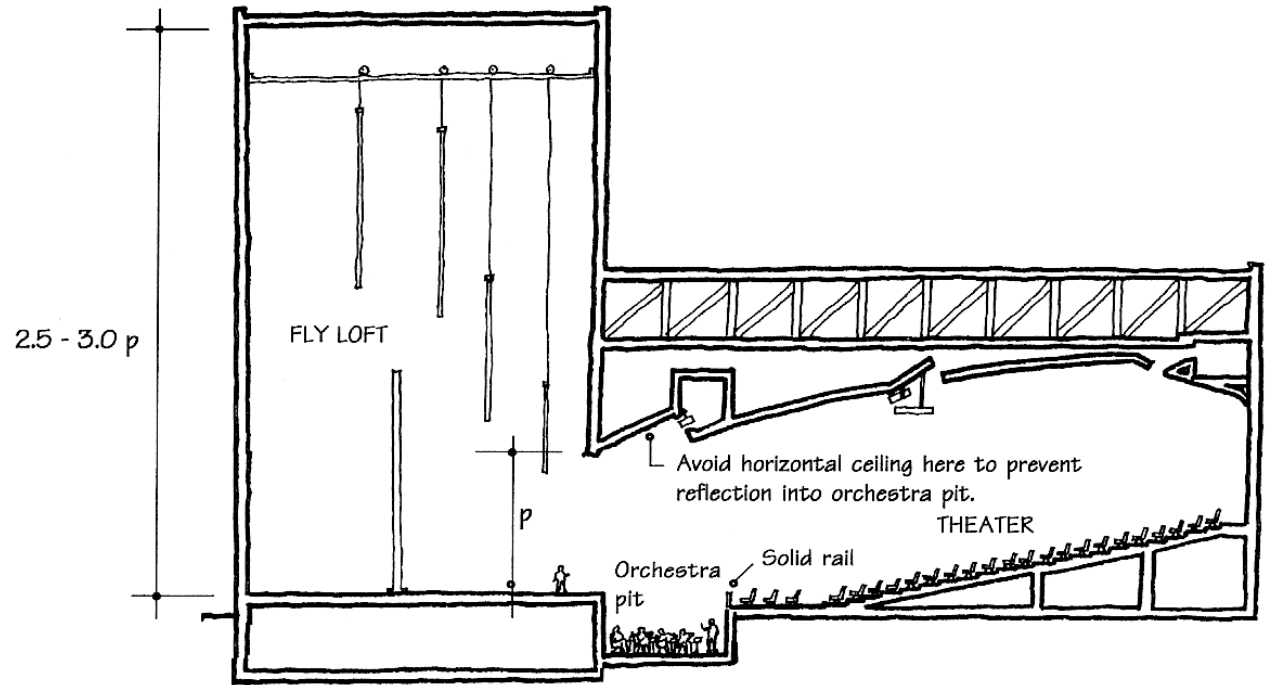
- Sources of ambient noise
 - External noise: traffic and other exterior noise
 - Noise from adjoining rooms/spaces
 - Audience footsteps
 - Mechanical and electrical noise
- Use buffer spaces (e.g. corridors and storage) around hall
- Use sound/light locks for hall entries
- Carpet aisles to reduce footfall sound
- RC or NC 25 maximum recommended for HVAC background sound and RC/NC 20 for better halls

DRAMA THEATERS

- A *proscenium stage* (a “framed” stage) is common
- Speech and music make up a typical performance, so acoustics must be a compromise between the two with speech requirements dominating.



DRAMA THEATER



- Ceiling reflectors
- Stage enclosure
- Rear wall treatment
- Parallel surfaces/flutter echo
- Room shape & volume
- Intimacy/distance from stage
- Balcony depth and underside shaping
- Balcony surface treatment
- Orchestra pit
- Background HVAC noise
- Sound reinforcement system
- Floor rake & sight lines
- Sound/light locks
- Stage height
- Site considerations
- Sound isolation
- Wall construction

DRAMA THEATERS

A few new design considerations:

- **Reverberation time:** varies from 0.9-1.1 sec for small theaters up to 1.2 sec for larger rooms (Egan p. 64 & 133)
- **Stage enclosure:** the RT should be approximately the same as the hall RT
- **Orchestra pit:** include removable sound absorbing panels or variable absorption for flexibility. A solid rail at the top of the pit will block excessive sound in the front row.
- **Seating:** Mehta says design with seats that have roughly the same absorption with or without the audience. Other designers say that people always add absorption so the seats should not be too absorptive to start with. Different design philosophies.

MUSIC PRESENTATION

- Music spaces generally have longer reverberation times than spaces for speech only
- Five design considerations are the same as speech rooms but may have different methods to obtain optimal design
 - Optimize reverberation time
 - Eliminate acoustical defects such as echoes and flutter echoes
 - Maximize loudness in audience
 - Minimizing the ambient noise level
 - Provide speech reinforcement system where needed

CONCERT HALLS

- Many of the same design features of drama theaters apply to concert halls.
- The reverberation time of a hall for music must be longer than a hall for speech.
 - Since opera is a combination of the two, its optimal RT is between speech and music.
 - Organ halls generally have a longer reverberation time when it does not interfere with other hall uses

CONCERT HALLS

1. Determine the room volume

- Rectangular halls - 300 ft³/person
- Surround (arena) hall - 450 ft³/person

2. Gross shaping of hall

- architect should study highly rated halls in detail.
- Narrow rectangular (“shoebox”) and horseshoe shaped rooms have worked best for orchestral music.
- Fan shapes take very careful design of reflective surfaces to be successful.

3. Will hall have balconies and box seating?

- Balcony depth (maximum depth is same as under balcony opening height) and balcony underside shaping are then considerations to avoid acoustical shadows

4. Diffusive or absorptive treatment should be used on rear wall and balcony face to reduce spectral reflections.

5. Balcony faces can be diffusive, absorptive or acoustically transparent.

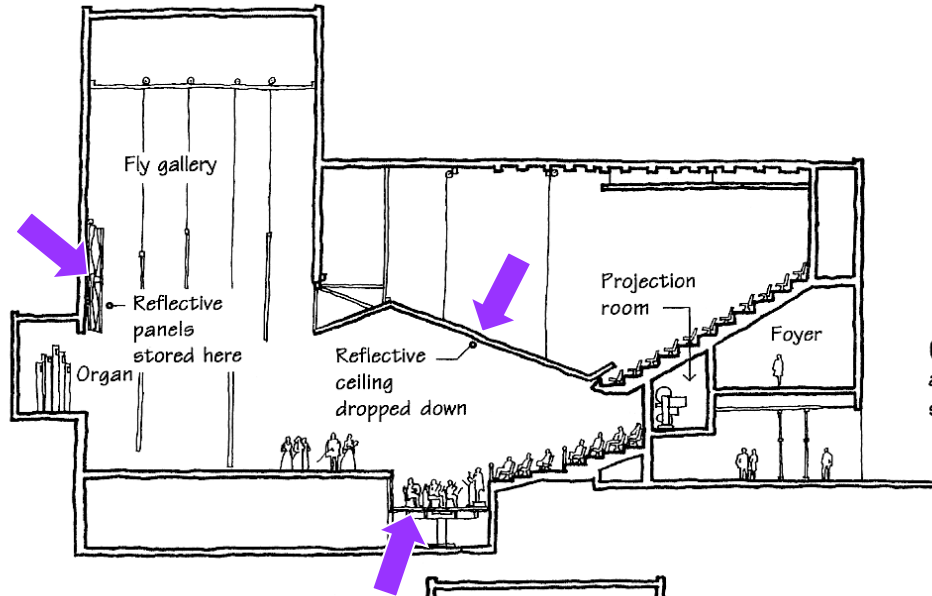
CONCERT HALLS

5. **Diffusive treatments on other surfaces** provide an “enveloping” sound field within the space, which is desirable. Complex surfaces (convex or angled) made of heavy, stiff materials such as fiberglass-reinforced gypsum can be mass produced.
6. **Interior finishes** are typically masonry or plaster on masonry. Avoid thin panels (wood, plaster, drywall, plastic) with air spaces behind them. They act as panel absorbers and reduce low frequency RT.
7. **Background sound levels** due to HVAC should be RC/NC 15 to 20 in the best halls and RC/NC 25 to 30 in “budget” halls. This means LARGE ducts for slow air velocities.
8. **Sound amplification systems**
 - Central loudspeaker cluster system – multiple loudspeakers grouped at one central location in a space – preferable for concert hall (as opposed to distributed loudspeaker system)

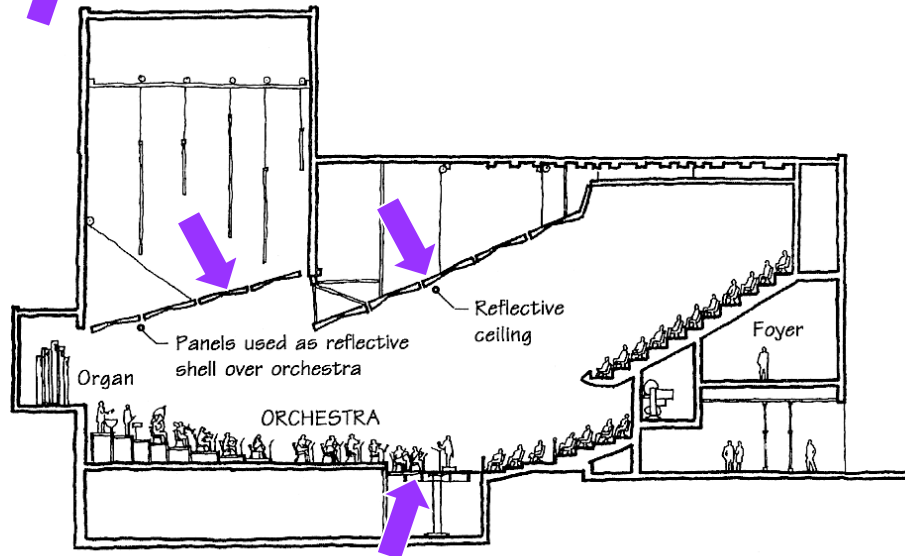
MULTIPURPOSE HALLS

- Multifunction halls accommodate several functions, such as concert hall, drama theater and lecture hall, and/or banquet facility
- Worship spaces can be considered multi-purpose since services typically consist of speech and music
- Acoustical compromise is required
- Determine the primary use and provide adjustments for other uses

MULTIPURPOSE HALLS



(a) THEATER MODE —
an intimate space with a
short reverberation time

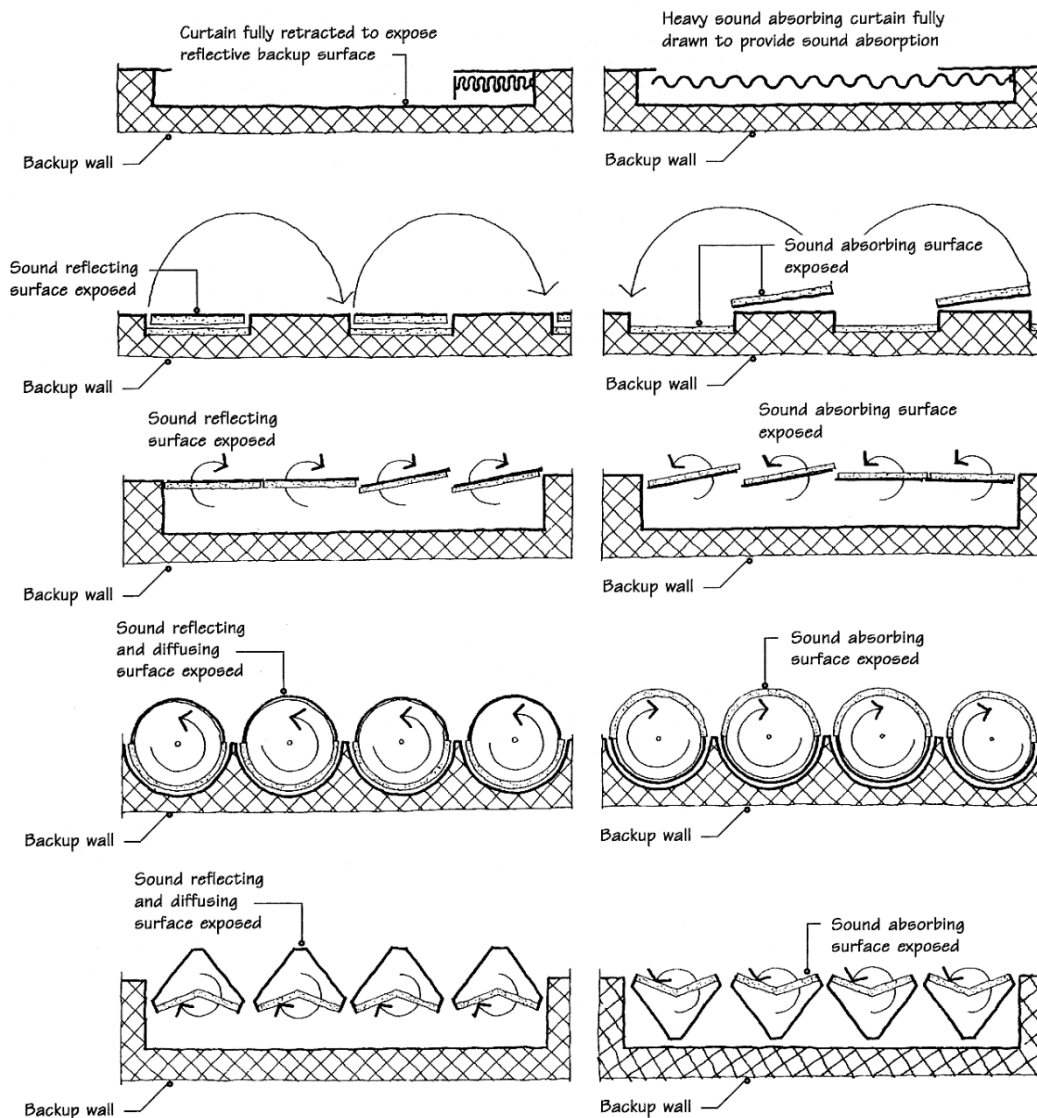


(b) CONCERT HALL MODE —
larger space with a long
reverberation time

Changes to
surfaces, room
volume, seating
arrangements
possible

VARIABLE ABSORPTION

Variable sound absorption allows for surfaces to be altered from reflective, to absorptive to diffusive to adjust for speech or music use.



VARIABLE ABSORPTION

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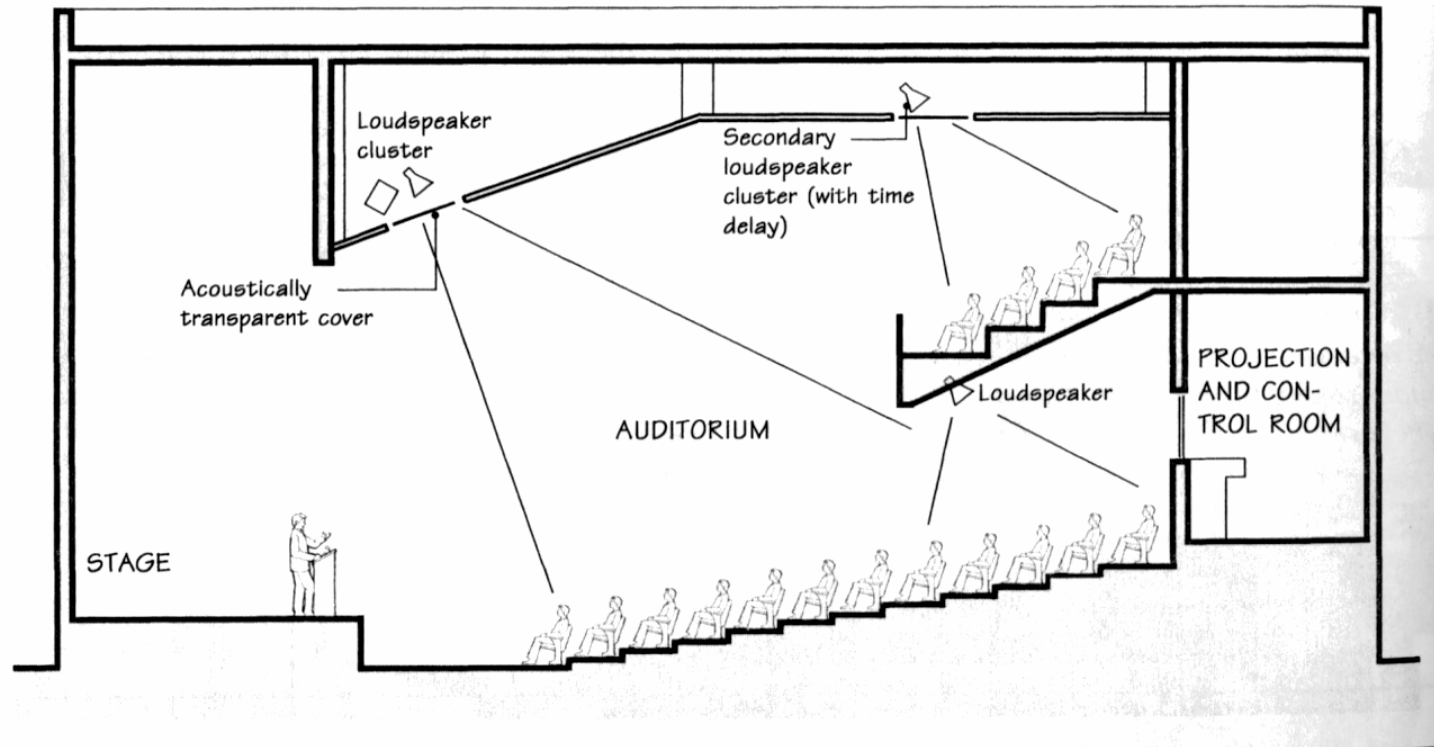
VARIABLE ABSORPTION

- Acoustical panels (black) slide from behind reflective convex covers



SOUND REINFORCEMENT

- Performance auditoria typically use a center cluster loudspeaker arrangement for the house system with other loudspeakers as needed for full audience coverage



AUDIO/VISUAL CONSIDERATIONS

Support spaces for performance halls with sound reinforcement and video systems

- **Amplifier Room** – Amplifiers get very hot and are noisy, so they are contained in a **separate space for cooling and noise**. It is best to locate them close to the loudspeakers so costly cable runs are reduced in length.
- **Dimmer Room** – similar in nature to an Amplifier Room. It is least expensive to locate them close to the theatrical lighting.
- **Post-Production Editing or Recording Rooms** – The need for these spaces depends on the occupants needs.
 - High TL & low RT similar to control room
- **Central Equipment Room** – for facilities where centralization of equipment is desired

CINEMAS

- The sound amplification system does the job of amplification in a cinema, so the wall surfaces do not have to be reflective to project sound.
- All the acoustical effects are included on the soundtrack
- To avoid coloration of the sound by the room characteristics, the room sound be acoustically dead (absorptive), which means a low reverberation time.

CINEMAS

- **Sound absorbing surfaces:** entire rear wall, wall behind screen and wall surrounding screen must be absorptive, and the walls and ceiling treated to achieve design RT
 - 2" to 4" fiberglass panels furred off wall 4" is usually necessary to obtain low frequency design RT
 - Carpet floors to reduce foot fall sound.
- **Sound isolation:** very important in multiple screen facilities. STC > 65, sound locks
- **Shape, volume and floor rake:** minimize volume, eliminate flutter echo, "stadium" seating
- **Background noise levels:** NC/RC 25-30
- **Projection room:** high STC partitions to block projection noise from hall

MUSIC PRACTICE ROOMS

- Spaces of various sizes are needed for various size groups (individual, small ensemble, full band, orchestra or chorus)
- Design considerations:
 - Reverberation time: 0.4 sec for individual practice room, band room around 0.65 sec, orchestra or choir room around 0.85 sec
 - Instrumental Room: 25 ft²/person and 400-650 ft³/person
 - Choir Room: 20 ft²/person and 400-530 ft³/person
 - Slant walls to control flutter echo
 - High sound insulation (partitions STC 60, doors STC 35 min. and preferably higher if near other rehearsal rooms or practice space) – Use full height walls, double walls and barrier ceilings as needed to meet high TL.
 - Sound absorption provided by acoustical ceiling and panels high in space. Diffusers can be mounted in ceiling or on walls.
 - Background levels of NC/RC 20-25
- Prefabricated individual practice rooms are available

MUSIC PRACTICE ROOMS

acoustical
diffusers

variable
absorption

CMU (split
face block
better
than flat)

acoustical
lay-in tile
ceiling

carpet?
May not
always be
desirable.



HOSPITAL CONSIDERATIONS

MAJOR ISSUES:

- Speech privacy
 - Doctor to patient/family discussions – individual consultation rooms
 - Medical (doctor/nurse/technician) staff conversation
 - Patient to visitor conversations
 - Achieve using:
 - High composite STC interior walls, doors, windows, etc., and effective partitions
 - Absorption to reduce sound propagation and noise levels
 - Higher ambient levels in corridors, waiting rooms, lobbies – RC/NC 40
 - Sound masking where needed

HOSPITAL CONSIDERATIONS

MAJOR ISSUES:

- Patient comfort
 - Quietness for resting/sleeping
 - Site location – away from loud exterior noise sources (e.g., airports, railroads, etc.)
 - Mechanical noise
 - Locate HVAC noise and vibration sources away from patient rooms – predict levels early
 - Floor planning - elevators, ward stations, janitor closets, etc. away from patient rooms
 - Power generators, transformers, etc.
 - Sound isolation
 - High composite STC exterior and interior walls, doors, window, etc.
 - RC/NC 30 – quiet but has some masking effect

Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4